

CAREX – Comprehensive Adaptive Rehabilitation Expert AI-Powered Personalized Rehabilitation Platform

PHASE I REPORT

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ANNA UNIVERSITY, CHENNAI
BONAFIDE CERTIFICATE

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ABSTRACT

CAREX (Comprehensive Adaptive Rehabilitation Expert) is an AI-powered personalized rehabilitation platform developed to support patients during their recovery process. Traditional rehabilitation often depends on periodic clinical visits and provides limited monitoring of patients while exercising at home. CAREX addresses this gap by providing personalized rehabilitation plans based on the patient's condition, recovery stage, and treatment history.

The system uses computer vision and MediaPipe Pose to monitor exercises, analyze body movements, evaluate exercise form, and count repetitions. It also uses AI-based recovery analysis to track patient progress and provide insights into recovery outcomes. Healthcare professionals can remotely review patient progress through a dedicated dashboard and modify rehabilitation plans when required. Additional features such as medication and appointment reminders support better adherence to the rehabilitation process. An optional ESP32 and MPU6050-based wearable module can assist with movement and fall detection.

CAREX aims to provide a continuous, personalized, and technology-assisted rehabilitation experience while supporting healthcare professionals in monitoring patients remotely.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CAREX	Comprehensive Adaptive Rehabilitation Expert
DFD	Data Flow Diagram
ESP32	Espressif Systems 32-bit Microcontroller
IMU	Inertial Measurement Unit
ML	Machine Learning
MPU6050	Motion Processing Unit 6050
ROM	Range of Motion
SpO₂	Peripheral Oxygen Saturation
UML	Unified Modeling Language
UI	User Interface
XGBoost	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Rehabilitation is an important part of recovery for patients who have undergone surgery, experienced fractures, sports injuries, stroke or other physical conditions. Recovery does not depend only on medical treatment; patients also need to perform prescribed exercises regularly, follow treatment instructions and attend scheduled consultations. However, maintaining this process outside the clinical environment can be difficult because healthcare professionals cannot continuously monitor every patient's activities.

In many cases, patients perform rehabilitation exercises at home without immediate feedback regarding their posture, movement or consistency. Progress may only be reviewed during periodic consultations, making it difficult to identify problems at an early stage. Patients may also forget medications, appointments or scheduled exercises, which further affects the overall outcome of the rehabilitation process.

With the development of Artificial Intelligence, computer vision, machine learning and cloud-based healthcare technologies, it is possible to provide more continuous support during rehabilitation. CAREX (Comprehensive Adaptive Rehabilitation Expert) is proposed as an AI-powered personalized rehabilitation platform that connects patients and healthcare professionals through a unified digital environment, using patient-specific information such as medical condition, age, recovery stage and treatment history to support personalized rehabilitation and recovery tracking.

1.2 OBJECTIVE

The main objective of this project is to develop an AI-powered rehabilitation platform that provides personalized, continuous and data-driven rehabilitation support while allowing healthcare professionals to remotely monitor patient progress. The specific objectives of the system are:

- To create a personalized rehabilitation profile for every patient.
- To provide rehabilitation exercises according to the patient's condition and recovery stage.

- To evaluate exercise performance using computer vision.
- To count exercise repetitions automatically and identify incorrect exercise posture or form.
- To track rehabilitation progress over time, including range of motion, adherence and pain trend.
- To monitor relevant patient health information such as heart rate and SpO₂.
- To use machine learning for recovery and readiness analysis.
- To provide medication, hydration and appointment support.
- To provide an AI-powered rehabilitation assistant for patient queries.
- To allow doctors and healthcare professionals to monitor patients remotely and modify treatment plans.
- To provide reports and useful recovery analytics to support clinical decision-making.

1.3 EXISTING SYSTEM

The existing rehabilitation process can generally be divided into the following approaches.

1.3.1 In-clinic Physiotherapy

The patient visits a hospital or clinic and the physiotherapist observes the exercises directly. Supervision is available only during the scheduled appointment, and the physiotherapist has no visibility into how the patient performs exercises between visits.

1.3.2 Paper-based or Verbal Instructions

The patient receives exercise instructions on paper or verbally and performs them at home. There is no automatic feedback about whether the exercise is being performed correctly, and adherence depends entirely on the patient's own discipline and memory.

1.3.3 General Fitness Applications

Fitness applications can track workouts, repetitions and general activity levels. However, they are not designed around a patient's specific injury, medical history or rehabilitation stage, and therefore cannot substitute for a rehabilitation-specific system.

1.3.4 Telehealth / Video Consultation

A patient can communicate with a physiotherapist through a video call, which provides remote interaction. However, this is still largely periodic, depends on both parties being available at the same time, and does not continuously collect rehabilitation information between consultations.

Limitations of Existing Systems

- Periodic, appointment-based monitoring rather than continuous supervision.
- No automatic, real-time feedback on exercise form or technique.
- Limited personalization to the patient's specific condition and recovery stage.
- No integrated recovery prediction or readiness analysis.
- Poor adherence monitoring for exercises, medication and appointments.
- Separate, disconnected tools for different rehabilitation-related activities.

1.4 PROPOSED SYSTEM

CAREX is proposed as an AI-powered personalized rehabilitation platform that integrates patient assessment, rehabilitation planning, exercise monitoring, progress tracking, recovery analysis and healthcare-professional supervision within a single system. Rather than treating rehabilitation as a set of disconnected tools, CAREX brings together a patient application, an AI/processing engine and a doctor dashboard into one coordinated platform.

The system creates a personalized rehabilitation profile for each patient based on their medical condition, age, recovery stage and treatment history. During exercise sessions, a computer-vision component analyzes the patient's body movement through a smartphone camera, calculates joint angles, counts repetitions and evaluates exercise form. This is complemented by an optional wearable extension (ESP32 with an MPU6050 IMU) for situations where camera-based measurement may be affected by lighting or occlusion.

Rehabilitation progress is tracked using indicators such as range of motion, adherence rate, form compliance and pain trend. A machine-learning component, built around XGBoost, supports recovery analysis by estimating recovery duration, identifying slow-recovery risk and suggesting the appropriate next rehabilitation stage. An AI rehabilitation assistant helps patients with day-to-day queries, while medication, hydration and appointment reminders support overall adherence. Throughout, the AI

components are designed to provide analytical support and recommendations, while healthcare professionals remain responsible for reviewing and modifying the treatment plan.

The overall proposed flow of the system can be summarized as:

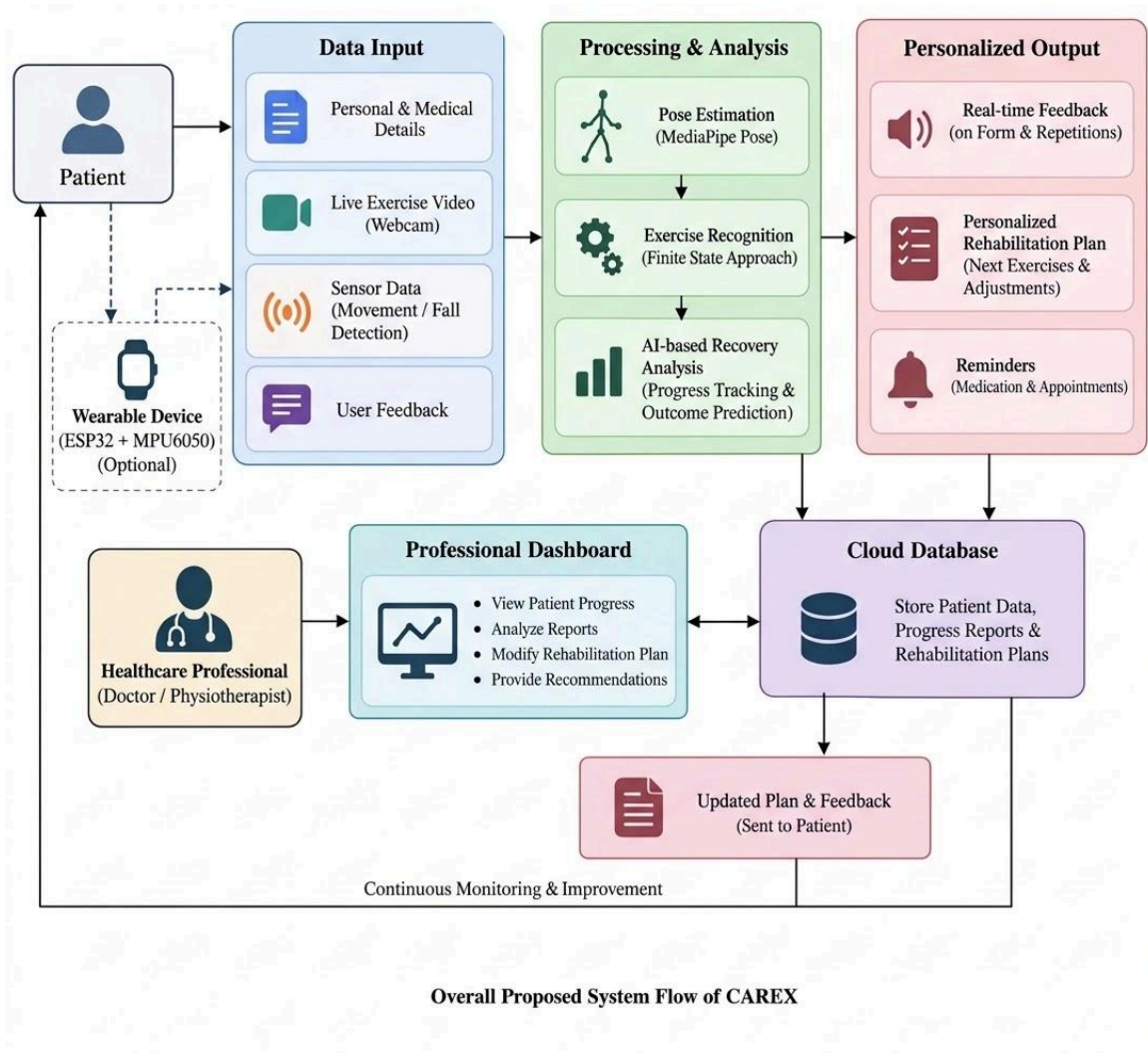


Fig. 1.1 Overall Proposed System Flow

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

2.1.1 Telerehabilitation for Stroke Patients

Laver et al. studied telerehabilitation for stroke patients and found that remote rehabilitation can be useful and can produce outcomes comparable to conventional, in-clinic rehabilitation in some situations. However, the approach does not by itself provide the complete personalization and continuous intelligent analysis that CAREX aims to provide.

2.1.2 AI-based Rehabilitation Technologies

Sumner et al. reviewed AI-based rehabilitation technologies, including mobile applications, robotics, gaming and wearable devices. The review also identified challenges such as reliability, the technology skills required of patients, user fatigue, and limited clinical evaluation of these systems.

2.1.3 Personalized Exercise Recommendation

Kim et al. investigated personalized exercise recommendation using a graph neural network, showing that different patient characteristics can be used to recommend more suitable exercises. However, the approach focuses mainly on recommendation rather than complete rehabilitation management.

2.1.4 Machine-Learning-based Recovery Prediction

Campagnini et al. studied machine-learning methods for predicting functional recovery after stroke, supporting the idea that ML can be useful for recovery prediction. The literature in this area also highlights problems such as small datasets, limited external validation, and differences between datasets and variables used across studies.

2.1.5 Digital Exercise Applications

Davergne et al. studied digital applications that provide personalized exercise videos, demonstrating the usefulness of digital applications for exercise-based rehabilitation. However, such exercise applications alone do not necessarily provide continuous monitoring, recovery prediction, doctor supervision or treatment management.

2.1.6 Mobile Health Applications

Nussbaum et al. reviewed mobile health applications used in rehabilitation and found that mobile health can improve accessibility and patient engagement. However, effectiveness depends on usability, clinical integration and how well the application supports the overall rehabilitation process.

2.1.7 AI-based Exercise Assessment

Li et al. developed a CNN-based system for chronic stroke assessment and exercise recommendation, showing that AI can assist rehabilitation assessment. The approach, however, is focused on a specific condition and particular assessment tasks rather than acting as a general rehabilitation management platform.

2.1.8 Conversational AI in Healthcare

Li et al. reviewed AI conversational agents in healthcare and noted that conversational AI can support patients, but it should complement professional supervision rather than replace it — a principle that directly informs the design of the CAREX rehabilitation assistant.

2.1.9 ReWin Digital Rehabilitation Solution

A pair of studies on the ReWin digital rehabilitation solution demonstrate how structured digital rehabilitation systems can support patients with stroke-related disabilities in India. Their focus, however, is more condition-specific than the integrated, multi-condition approach that CAREX aims to provide.

2.2 COMPARISON AND DISCUSSION

The table below compares the major existing approaches to rehabilitation support against the CAREX approach.

Existing Approach	Main Capability	Major Limitation	CAREX Approach
In-clinic physiotherapy	Professional supervision	Periodic monitoring	Continuous digital monitoring
Paper / verbal instructions	Exercise guidance	No feedback	Interactive exercise support
Fitness applications	Activity tracking	Not rehabilitation-specific	Patient-specific rehabilitation
Telehealth	Remote consultation	Limited between-session monitoring	Continuous progress information

Existing Approach	Main Capability	Major Limitation	CAREX Approach
Exercise recommendation systems	Personalized exercises	Limited overall management	Personalized rehabilitation workflow
Recovery prediction models	Recovery analysis	Focused prediction task only	Prediction integrated with rehabilitation
AI assistants	Conversational support	Cannot replace professionals	Assistant + professional supervision
CAREX (proposed)	Integrated rehabilitation support	Requires further clinical validation	Personalized + AI + monitoring + doctor involvement

Table 2.1 Comparison of Existing Approaches with CAREX

2.3 CONCLUSION

The literature survey shows that artificial intelligence, telerehabilitation, personalized exercise recommendation, machine learning and conversational systems can each contribute to rehabilitation. However, many existing solutions focus on only one or two aspects of the recovery process. CAREX therefore combines personalized rehabilitation planning, exercise monitoring, progress tracking, recovery analysis, patient support and healthcare-professional supervision within a unified platform.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The CAREX platform is organized into five logical layers, described below.

Layer 1 — Patient Interface

Handles registration, patient profile, rehabilitation plan display, the exercise screen, health information, reminders and the AI assistant.

Layer 2 — Data Collection

Collects patient profile data, exercise information, progress data, health information and rehabilitation history.

Layer 3 — AI / Processing

Performs computer-vision-based exercise analysis, repetition counting, form evaluation, ML-based prediction / readiness analysis, and powers the AI assistant.

Layer 4 — Database / Cloud

Stores patient data, rehabilitation records, exercise session data, progress information and health information.

Layer 5 — Doctor Dashboard

Provides patient monitoring, progress analytics, reports and treatment-plan review / modification for healthcare professionals.

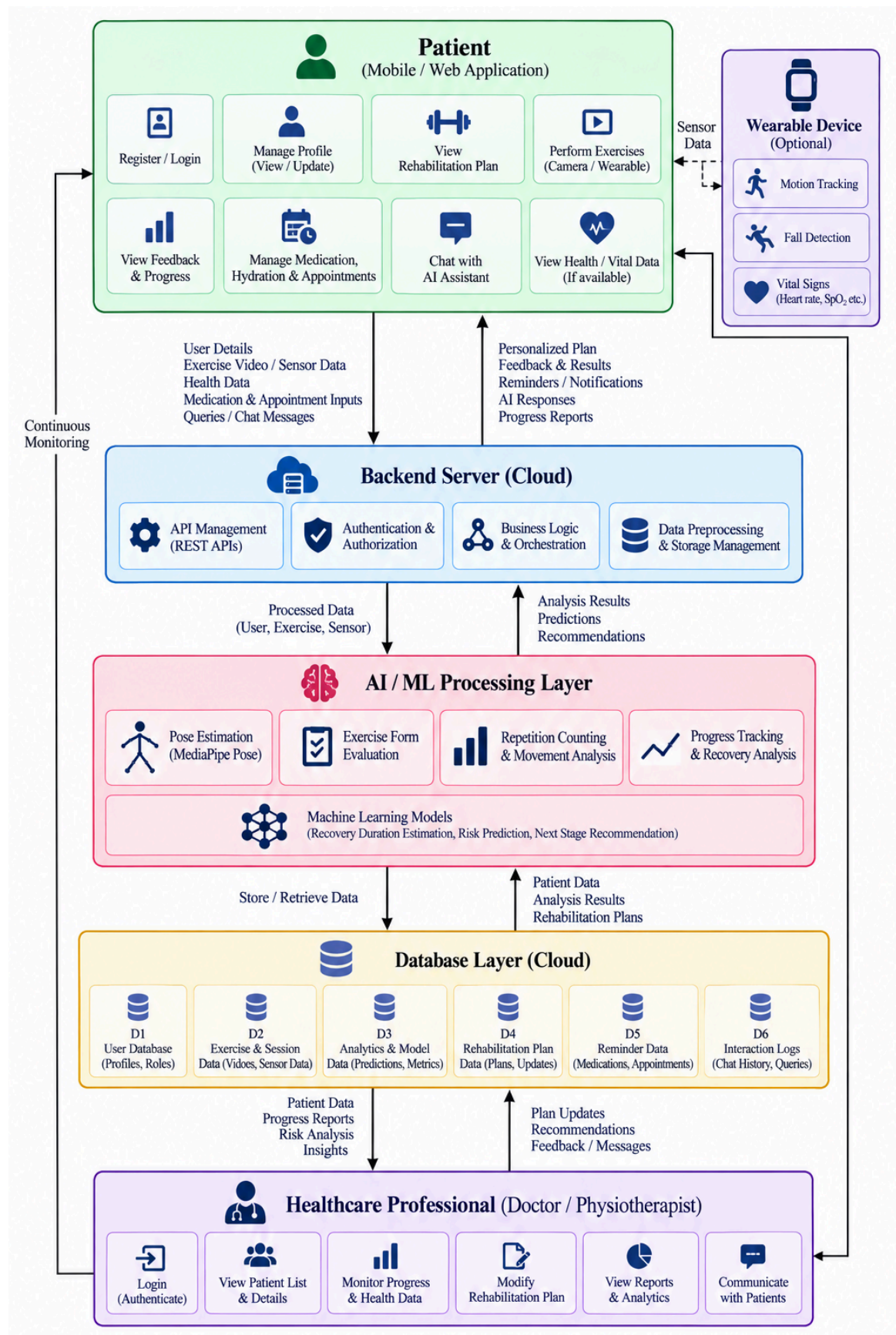


Fig. 3.1 CAREX System Architecture

Figure 3.1 illustrates the system architecture of CAREX and the interaction between its major components. The Patient interface collects profile, rehabilitation, exercise, and health-related information, while the optional wearable device provides sensor data such as movement and fall detection. The backend server manages authentication, business logic, data processing, and communication between the system components. The AI/ML layer performs pose estimation, exercise-form evaluation, repetition counting, progress tracking, and recovery analysis. The database stores patient, exercise, analytics, rehabilitation-plan, reminder, and interaction data. The Healthcare Professional dashboard allows professionals to monitor patient progress, review reports and insights, modify rehabilitation plans, and communicate with patients.

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

- Mobile / frontend application framework for the patient and doctor interfaces.
- Backend / API service layer for business logic and orchestration.
- Firebase for authentication, role-based access and real-time profile data.
- Computer-vision library (MediaPipe Pose) for pose landmark extraction.
- Machine-learning libraries for the XGBoost-based recovery/readiness model.
- Cloud / database services for persistent storage of rehabilitation records.

3.2.2 Hardware Requirements

Basic:

- Laptop / desktop for development and the doctor dashboard.
- Smartphone or device with a camera for exercise monitoring.
- Internet connectivity for cloud synchronization.

Optional:

- ESP32 microcontroller.
- MPU6050 6-axis IMU (accelerometer + gyroscope).

The ESP32 + MPU6050 combination forms an optional wearable extension, used only in situations where camera-based measurement may be affected by lighting or occlusion; it is not a mandatory part of the core system.

3.2.3 Data Requirements

Patient information: age, medical condition, injury / condition, recovery stage, treatment history.

Rehabilitation information: prescribed exercises, repetitions, sessions, exercise performance, progress.

Health information: relevant vitals, pain information and other patient-entered health data.

AI information: exercise metrics, recovery metrics, and readiness / recovery predictions.

3.2.4 Security and Privacy

Because CAREX handles healthcare-related data, the design includes authentication, role-based access control, controlled patient/doctor access boundaries, secure storage and protection of patient information and privacy. Data security, privacy and clinical validation are treated as important considerations to be strengthened as the system matures.

3.3 SYSTEM DESIGN

3.3.1 Use Case Diagram

The two primary actors of the system are the Patient and the Doctor.

- Patient: Register / Login, Complete Profile, View Rehabilitation Plan, Perform Exercise, View Exercise Feedback, View Progress, Track Health, Manage Medication / Hydration, Ask AI Assistant.
- Doctor/Health care Professional : Login, View Patients, View Progress, View Reports, Review AI Analysis, Modify Treatment plan.

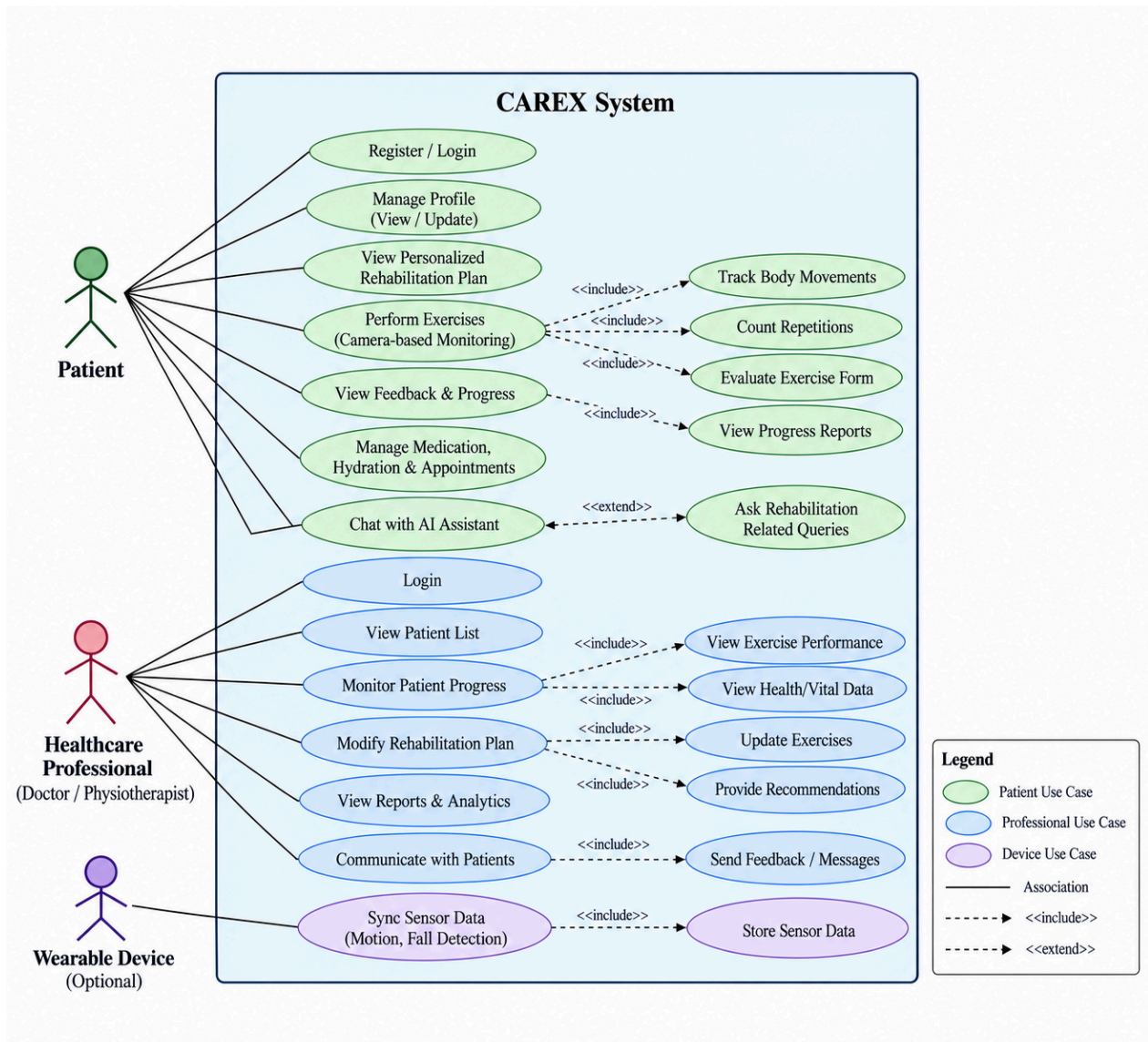


Fig. 3.2 Use Case Diagram

Figure 3.2 illustrates the major interactions between the actors and the CAREX system. The Patient can register and manage their profile, view the personalized rehabilitation plan, perform exercises with camera-based monitoring, view exercise feedback and progress, manage medication, hydration and appointments, and interact with the AI assistant. The Healthcare Professional can log in, view patient details and progress, review reports and AI-based insights, modify rehabilitation plans, and communicate with patients. The optional Wearable Device provides sensor data such as movement and fall detection to the system. The diagram shows how these actors interact with the different functionalities of CAREX.

3.3.2 Activity Diagram



Fig. 3.3 Activity Diagram

Figure 3.3 represents the sequence of activities involved in the CAREX rehabilitation process. The process begins with patient registration and profile creation, followed by the generation of a personalized rehabilitation plan. The patient performs the prescribed exercises using camera or wearable-based monitoring, after which the system processes the collected data and analyzes exercise performance, repetitions, and recovery progress. Feedback and progress information are provided to the patient, while healthcare

professionals can monitor the results and modify the rehabilitation plan when required. The process also includes AI-assisted support and medication or appointment reminders to provide continuous assistance throughout rehabilitation.

3.3.3 Data Flow Diagram (Level 1)

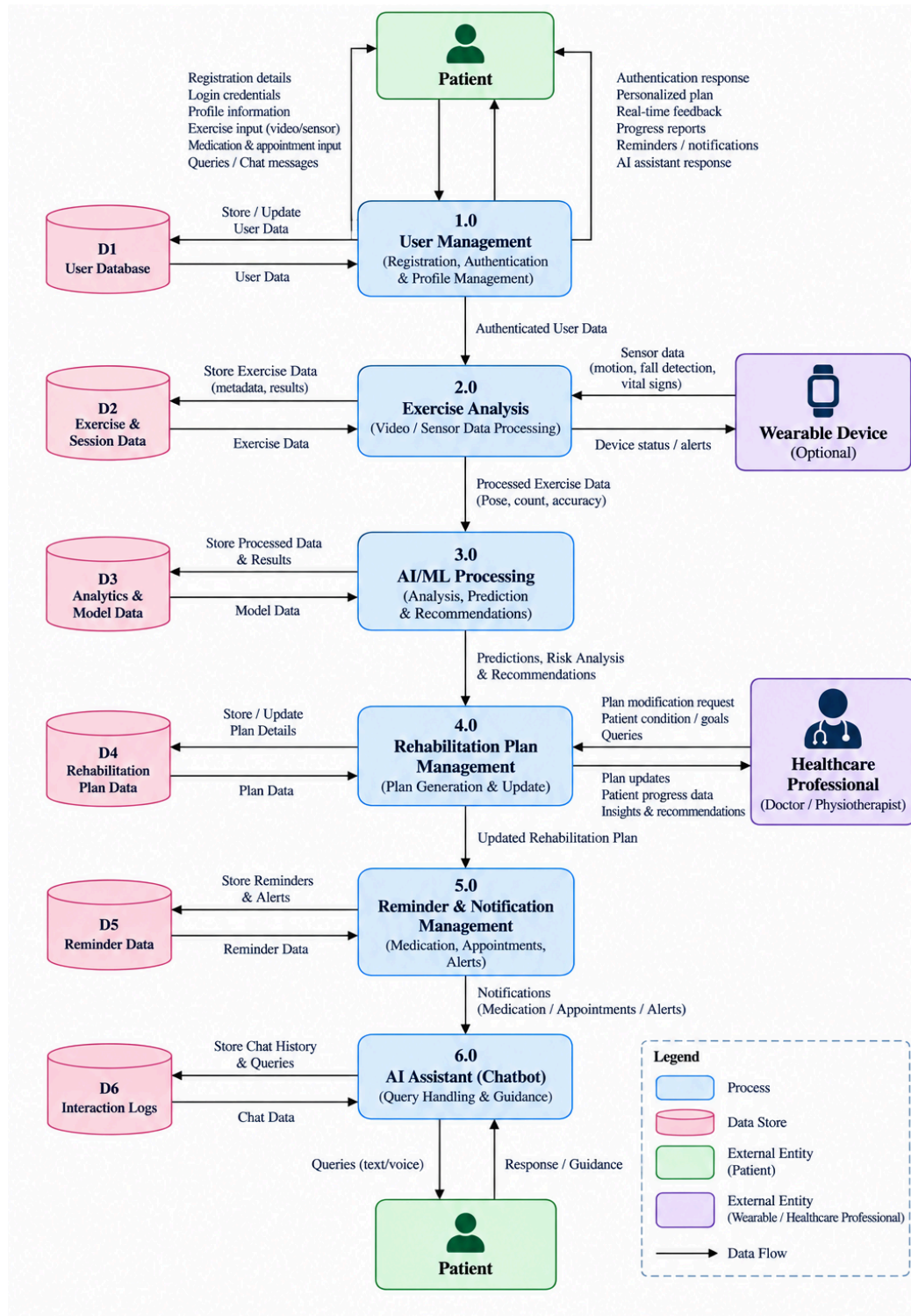


Fig. 3.4 Data Flow Diagram of the System

Figure 3.4 illustrates the Level 1 Data Flow Diagram of the CAREX system and shows how information moves between the patient, system processes, data stores, healthcare professional, and optional wearable device. The process begins with user registration, authentication, and profile management, after which patient and rehabilitation information is passed to the exercise and sensor data processing module. Exercise video and wearable sensor data are analyzed to obtain movement, repetition, form, and other relevant metrics. The AI/ML module uses the processed information to evaluate exercise performance, track progress, and support recovery analysis and recommendations. These results are used by the rehabilitation plan management module to generate or update personalized plans, while the reminder and notification module manages medication, appointment, and exercise-related reminders. The AI assistant handles patient queries and provides rehabilitation-related guidance. The healthcare professional can review patient progress, reports, and AI-generated insights and provide plan updates or recommendations. The respective data stores maintain user, exercise, analytics, rehabilitation-plan, reminder, and interaction information, allowing the system to support continuous and personalized rehabilitation.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

CAREX, which stands for Comprehensive Adaptive Rehabilitation Expert, is developed as an AI-powered personalized rehabilitation platform that integrates patient information, rehabilitation planning, exercise monitoring, progress tracking, recovery analysis and healthcare-professional supervision within a unified digital environment. The methodology followed in CAREX is designed around the continuous collection and analysis of patient-specific rehabilitation information so that the system can provide personalized support throughout the recovery process. The methodology begins with patient registration and profile creation, where information such as age, medical condition, injury or physical condition, recovery stage and treatment history is collected. This information is used to establish a personalized rehabilitation profile for the patient and forms the basis for recommending suitable rehabilitation activities.

After the patient profile is established, the rehabilitation plan is created according to the patient's condition and recovery stage. The patient can access the prescribed exercises through the application and perform them using a smartphone or camera-enabled device. During an exercise session, the computer-vision component captures the patient's body movement and uses MediaPipe Pose to identify relevant body landmarks. These landmarks are processed to understand movement patterns, estimate joint positions and angles, count repetitions and evaluate whether the exercise is being performed correctly. This allows CAREX to provide immediate exercise-related feedback rather than depending entirely on periodic physical consultations. The system architecture identifies exercise information and sensor information as important inputs to the processing layer, where movement, repetition and form-related metrics are obtained.

The processed exercise information is subsequently used for progress tracking and recovery analysis. CAREX considers indicators such as range of motion, exercise adherence, form compliance and pain trends to understand the patient's rehabilitation progress over time. The machine-learning component of the system is designed around XGBoost to support recovery and readiness analysis. The processed rehabilitation data

can be used to estimate recovery-related outcomes, identify possible slow-recovery conditions and support recommendations regarding the next stage of rehabilitation. However, the AI components are intended to provide analytical support rather than replace professional medical decisions. Healthcare professionals remain responsible for reviewing the patient's progress, evaluating AI-generated insights and modifying the rehabilitation plan whenever required.

The methodology also includes continuous patient support through reminders and an AI-powered rehabilitation assistant. Medication, hydration, appointment and exercise-related reminders are incorporated to improve adherence to the rehabilitation process, while the conversational assistant provides support for patient queries. At the healthcare-professional end, a dedicated dashboard allows doctors or physiotherapists to review patient information, exercise progress, reports and AI-generated insights and to update rehabilitation plans. The overall methodology therefore follows a continuous cycle of patient assessment, personalized planning, exercise execution, computer-vision-based analysis, progress tracking, recovery analysis and professional review. This approach allows CAREX to combine personalized rehabilitation with continuous digital monitoring while maintaining healthcare-professional involvement in treatment decisions.

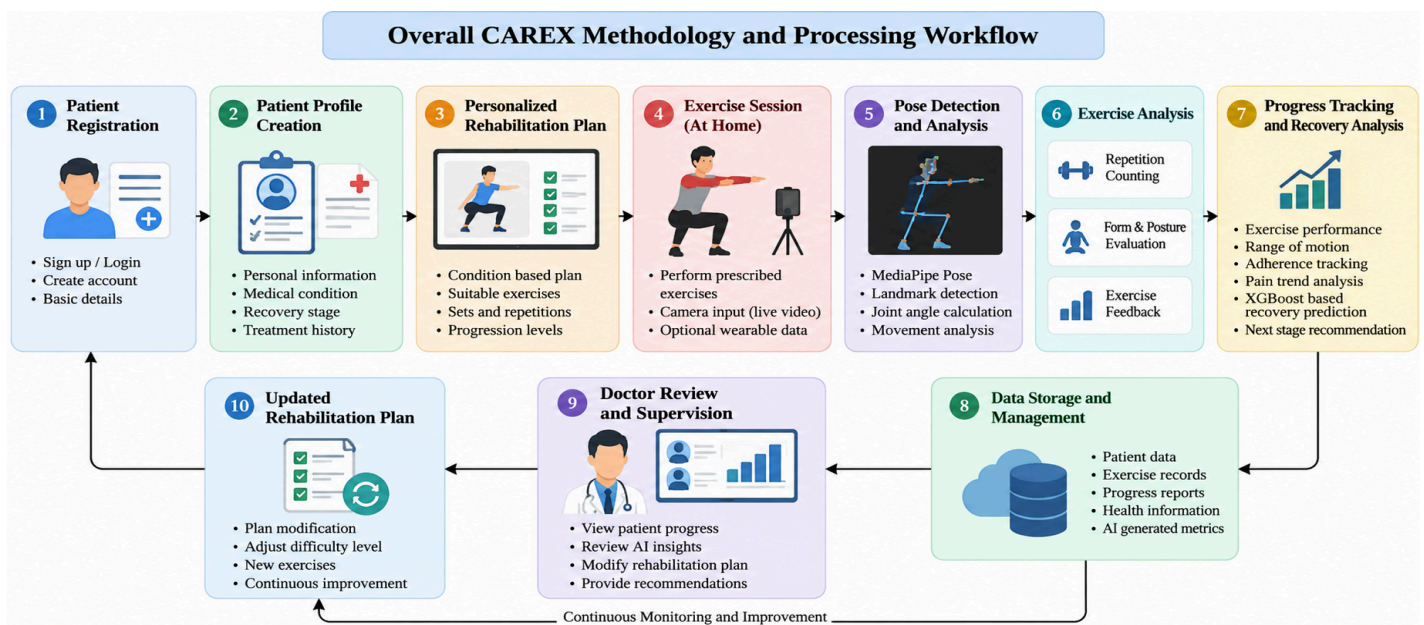


Figure 4.1: Overall CAREX Methodology and Processing Workflow

4.2 MODULES

The CAREX platform is organized into a set of functional modules that collectively implement the major activities involved in personalized rehabilitation. The modules are designed to separate patient management, rehabilitation activities, AI-based processing, professional supervision and continuous patient assistance. This modular organization allows individual components to be developed and improved independently while maintaining communication between them through the backend and database layers. The architecture of CAREX consists of patient-facing functionality, data collection, AI and processing components, cloud or database services and a healthcare-professional dashboard.

4.2.1 DATA MANAGER PACKAGE

The Data Manager Package is responsible for collecting, organizing and maintaining the information required by the CAREX platform. The module manages patient-specific information including age, medical condition, injury or physical condition, recovery stage and treatment history. It also manages rehabilitation information such as prescribed exercises, repetitions, sessions and exercise performance, along with relevant health information, pain information and AI-generated rehabilitation metrics. The collected information is validated and stored so that it can be accessed by other components of the system whenever required. The data management layer acts as the foundation for personalization because the rehabilitation plan and subsequent analysis depend on the availability of accurate patient and rehabilitation information.

The Data Manager Package also maintains information generated during exercise sessions. When a patient performs an exercise, the system generates movement-related information such as repetitions, form-related measurements and other exercise metrics. These results are associated with the corresponding patient and exercise session and can subsequently be used for progress tracking and recovery analysis. The database and cloud layer stores patient information, exercise records, rehabilitation plans, progress information, reminders and interaction data. By maintaining these records in a structured manner, CAREX can preserve the patient's rehabilitation history and provide healthcare professionals with a consolidated view of the recovery process.

4.2.2 CORE PACKAGE

The Core Package represents the central processing component of CAREX and is responsible for implementing the primary rehabilitation workflow. It receives information from the patient and data-management components and coordinates the execution of rehabilitation plans, exercise monitoring, progress analysis and AI-assisted processing. The core component connects the patient interface with the computer-vision, machine-learning and database components so that information generated during rehabilitation sessions can be processed and stored systematically. It therefore acts as the main coordination layer between the different functional components of the CAREX platform.

A major responsibility of the Core Package is exercise monitoring. During an exercise session, the patient's camera input is processed using MediaPipe Pose to identify body landmarks. The detected landmarks are used to analyse body movement and determine exercise-related parameters. Joint positions and movement patterns can be used to calculate relevant angles and identify movement transitions, which support automatic repetition counting and exercise-form evaluation. The results of this analysis are then made available to the patient as feedback and stored as part of the patient's rehabilitation history. CAREX specifically aims to evaluate exercise performance using computer vision, automatically count repetitions and identify incorrect posture or form.

The Core Package also coordinates progress tracking and recovery analysis. Exercise metrics collected over multiple sessions can be compared to identify changes in performance, adherence and range of motion. These processed features can then be supplied to the machine-learning component for recovery and readiness analysis. The architecture of CAREX places pose estimation, exercise-form evaluation, repetition counting, progress tracking and recovery analysis within the AI and processing layer, allowing these functions to work together as part of the rehabilitation workflow.

4.2.3 MODERATION PACKAGE

In CAREX, the moderation-related functionality is associated with maintaining the quality and appropriateness of information and recommendations presented to the patient. Since rehabilitation is a healthcare-related activity, the system must distinguish between analytical assistance provided by the AI components and actual clinical

decisions made by healthcare professionals. The moderation layer therefore supports the controlled presentation of AI-generated information and ensures that recommendations are treated as supportive information rather than independent medical decisions.

The moderation process is particularly important for AI-generated recovery insights and patient interactions with the rehabilitation assistant. Information generated from exercise analysis and recovery prediction can be presented to the patient in a controlled manner, while more significant changes to the rehabilitation plan remain under the supervision of healthcare professionals. The CAREX design explicitly maintains professional involvement in treatment decisions, with healthcare professionals able to review patient progress, examine AI-generated insights and modify rehabilitation plans when required. This approach provides a balance between automated assistance and professional supervision.

The moderation component can also support the handling of patient interactions with the AI assistant by ensuring that responses remain within the intended rehabilitation-support context. The assistant is designed to answer rehabilitation-related queries and provide day-to-day support, while clinical decisions remain with qualified healthcare professionals. This separation is important because CAREX is currently a proposed and developing system rather than a clinically validated replacement for professional rehabilitation services.

4.2.4 REINFORCEMENT PACKAGE

The Reinforcement Package in CAREX is associated with the continuous improvement of rehabilitation support using information collected from repeated patient interactions and exercise sessions. Rehabilitation is not a single-session activity; patient performance can change as recovery progresses. Therefore, the system maintains exercise history, progress information, adherence information and health-related observations so that subsequent rehabilitation decisions can be based on accumulated information rather than only the latest exercise session.

The package supports the adaptive nature of CAREX by allowing rehabilitation information to be reviewed over time. Exercise performance, range of motion, adherence and pain trends can contribute to understanding how the patient is progressing. These historical observations can be used by the AI and machine-learning components to

support recovery and readiness analysis. The resulting insights can then be reviewed by healthcare professionals, who can modify the patient's rehabilitation plan when necessary. In this way, the system forms a continuous feedback cycle in which patient activity generates data, the system analyses the data, and healthcare professionals use the resulting information to guide future rehabilitation activities.

The adaptive process is especially relevant to the XGBoost-based recovery analysis proposed in CAREX. The machine-learning component is intended to analyse rehabilitation-related features and support recovery-duration estimation, slow-recovery risk identification and next-stage recommendations. Since the current Phase-I system requires further validation using real clinical data, these outputs should be considered analytical support rather than clinically established predictions.

4.2.5 UI PACKAGE

The UI Package provides the interaction layer through which patients and healthcare professionals access the CAREX platform. The patient interface provides functions such as registration and login, profile management, rehabilitation-plan viewing, exercise execution, exercise feedback, progress monitoring, health tracking, medication and hydration management, appointment support and interaction with the AI rehabilitation assistant. The interface is designed to present the information required by patients in an accessible manner while allowing them to participate actively in their rehabilitation process.

The UI Package also includes the healthcare-professional interface. Through the doctor dashboard, healthcare professionals can log in, view patient information, monitor exercise progress, review reports and AI-generated insights, communicate with patients and modify rehabilitation plans. This creates two complementary interfaces within CAREX: one focused on patient participation and continuous rehabilitation support and another focused on professional monitoring and treatment-plan management. The user interface therefore acts as the final interaction layer between the underlying data, AI-processing components and the users of the system.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of CAREX follows the methodology and modular architecture described in the previous chapter. The implementation focuses on establishing the data-processing pipeline, exercise-monitoring functionality, AI-assisted analysis and user interaction components required for a personalized rehabilitation platform. The system processes patient information and rehabilitation data through different stages before presenting useful information to the patient or healthcare professional. The implementation is structured so that information generated during patient activity can be stored and reused for progress tracking and recovery analysis. The major implementation stages considered in this phase are data pre-processing, text processing where applicable, data representation, exercise analysis and AI-based rehabilitation analysis.

5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING

Data pre-processing is an important stage in CAREX because the quality of the information supplied to the analysis modules directly affects the usefulness of the resulting rehabilitation insights. The system receives different categories of information, including patient profile data, medical and rehabilitation information, exercise-session data, health information and AI-generated metrics. Before these data are used by subsequent modules, they need to be organized into a consistent structure. Patient information such as age, medical condition, recovery stage and treatment history is maintained as structured information, while exercise sessions are associated with the corresponding patient and prescribed exercise.

Exercise-monitoring data also undergo processing before being used for analysis. During a camera-based exercise session, the system receives video frames and extracts body landmarks using MediaPipe Pose. The detected landmarks provide the basis for calculating movement-related information and evaluating exercise performance. The

processing stage can therefore convert raw camera observations into meaningful rehabilitation metrics such as movement information, repetition counts and form-related measurements. These metrics can subsequently be stored as part of the patient's exercise history and used for progress tracking. CAREX identifies exercise performance, repetition counting and form evaluation as key functions of its computer-vision component.

The pre-processed information is also used for progress and recovery analysis. Exercise performance can be considered along with adherence, range of motion and pain trends to understand changes over multiple rehabilitation sessions. The processed information is then made available to the machine-learning component for recovery and readiness analysis. By maintaining structured and consistent information throughout the pipeline, CAREX creates a foundation for personalized rehabilitation analysis and future model improvement.

5.1.2 IMPLEMENTATION OF DATA TOKENIZATION

Tokenization is normally used when textual information has to be converted into smaller units that can be processed by a natural-language-processing model. In CAREX, textual information can arise from patient queries, AI-assistant interactions and other text-based rehabilitation information. The tokenization stage can therefore be considered part of the processing pipeline for the conversational assistant when textual input is supplied to an underlying language model. The purpose of tokenization is to transform the user's natural-language query into a machine-processable representation while preserving the meaning required for generating an appropriate response.

For example, when a patient enters a rehabilitation-related question, the input is first received by the AI-assistant interface and passed to the text-processing component. The text can then be divided into suitable tokens and converted into the representation expected by the underlying language model. The processed representation is subsequently supplied to the model for response generation. The resulting response is returned through the CAREX interface so that the patient can receive rehabilitation-related assistance. The Phase-I CAREX report establishes the presence of an AI-powered rehabilitation assistant for patient queries, but it does not specify a particular tokenizer or NLP architecture;

therefore, the exact tokenizer and model configuration should be documented after the final implementation is fixed.

The tokenization stage is therefore considered a supporting component of the conversational functionality rather than the primary rehabilitation-analysis pipeline. The major numerical and visual data used by CAREX for exercise analysis are obtained from patient information, exercise sessions and pose landmarks rather than from textual tokenization. This distinction is important because CAREX combines different forms of information processing, with computer vision being used for exercise monitoring and language processing being relevant primarily to the conversational assistant.

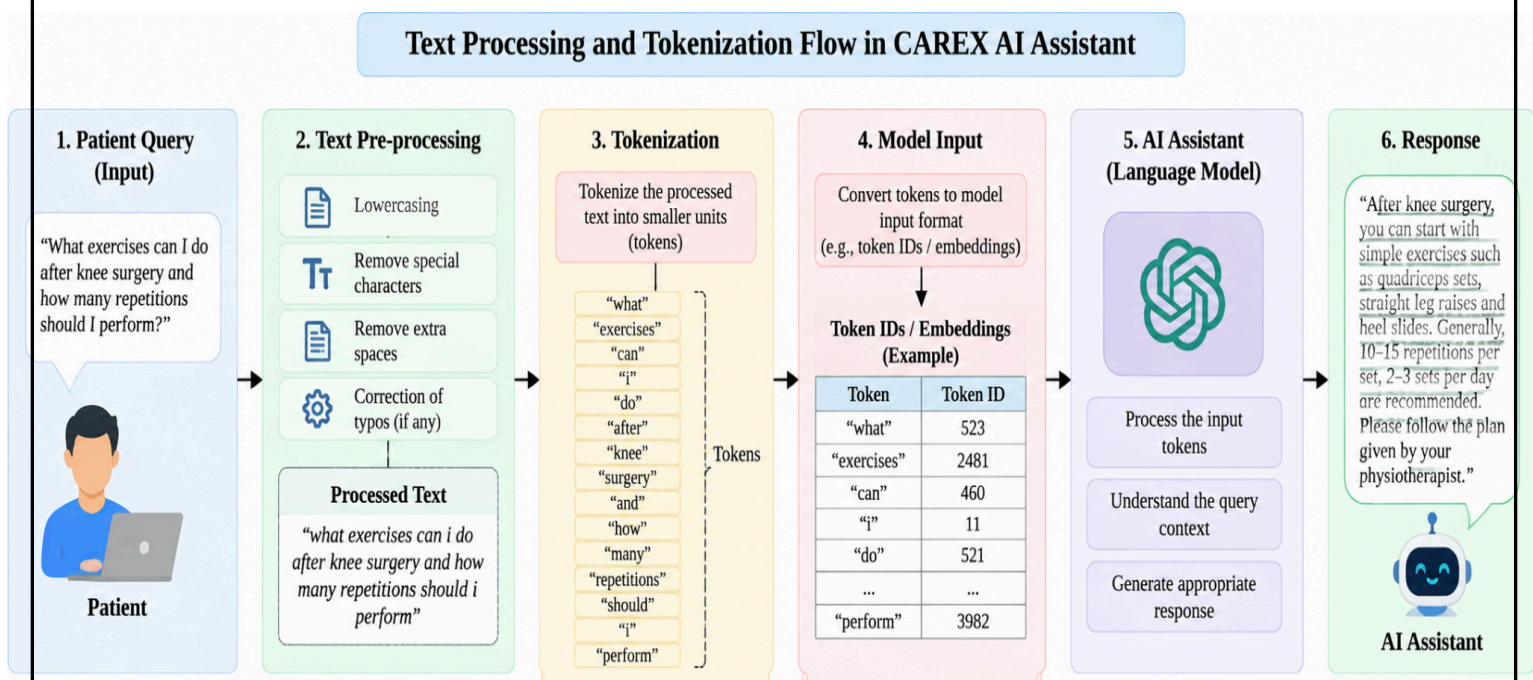


Figure 5.1: Text Processing and Tokenization Flow

5.1.3 IMPLEMENTATION OF DATA EMBEDDING

Data embedding refers to the conversion of processed information into numerical representations that can be used by machine-learning or artificial-intelligence models. Within CAREX, the requirement for numerical representation occurs in different processing components. Exercise-related information obtained from pose landmarks can be represented through numerical features such as joint coordinates, joint angles, movement measurements and repetition-related values. These numerical features provide

a structured representation of the patient's exercise performance and can be used for progress and recovery analysis.

The exercise-analysis pipeline begins with the extraction of body landmarks from camera frames using MediaPipe Pose. The landmark coordinates can be processed to derive movement-related features, including the relative positions of body joints and changes in joint angles during an exercise. These values form a numerical representation of the patient's movement and can be stored as exercise metrics. The resulting features can subsequently be combined with rehabilitation information such as adherence, pain trends and other relevant patient information for machine-learning-based recovery and readiness analysis. CAREX specifically proposes XGBoost for recovery and readiness analysis, making the preparation of meaningful numerical features an important part of the implementation pipeline.

For the conversational component, textual patient queries may also require numerical representation before being processed by the selected language model. However, the current Phase-I report does not identify a specific embedding model or embedding dimension, and therefore no particular embedding technique is claimed here. The exact embedding method can be documented in the subsequent implementation phase once the final AI-assistant architecture is established. The primary representation used by the current CAREX rehabilitation pipeline is the structured numerical representation of patient, exercise and movement-related information.

5.2 DISCUSSION

The implementation of CAREX demonstrates how artificial intelligence, computer vision, machine learning and digital healthcare components can be combined to create a unified rehabilitation-support platform. The major strength of the proposed implementation is that it does not treat exercise monitoring, progress tracking, recovery analysis and professional supervision as separate systems. Instead, information generated at one stage of rehabilitation can be stored and reused at later stages. Patient information is used to establish a personalized rehabilitation profile, exercise sessions generate performance information, and the resulting metrics contribute to progress and recovery analysis. This creates a continuous information flow throughout the rehabilitation process.

The computer-vision component provides an important part of this workflow because it enables the system to analyse exercises performed outside a clinical environment. MediaPipe Pose is used to extract body landmarks from camera input, allowing the system to analyse movement, count repetitions and evaluate exercise form. This provides a basis for giving patients feedback during exercise sessions and collecting structured exercise-performance information for later analysis. The CAREX design also considers an optional ESP32 and MPU6050-based wearable extension for movement and fall-related sensing when camera-based measurement is affected by lighting or occlusion.

The progress-monitoring component extends the usefulness of individual exercise sessions by maintaining information over time. Indicators such as range of motion, adherence, form compliance and pain trends can be used to understand whether the patient is progressing through the rehabilitation process. The XGBoost-based recovery-analysis component is intended to use rehabilitation-related information to support recovery-duration estimation, slow-recovery risk identification and readiness analysis. These predictions are intended to provide analytical assistance and should not be interpreted as replacements for clinical assessment. The healthcare professional remains responsible for reviewing patient information and making treatment-related decisions.

The healthcare-professional dashboard provides an additional layer of supervision by allowing doctors or physiotherapists to review patient progress, reports and AI-generated insights. This is particularly important because rehabilitation is highly dependent on individual patient conditions and recovery patterns. By allowing professionals to modify rehabilitation plans based on the information collected by the system, CAREX follows a human-supervised AI approach rather than a fully autonomous treatment system. The AI components therefore act as decision-support mechanisms while clinical responsibility remains with the healthcare professional.

The current implementation also has limitations that must be considered when interpreting the results. CAREX is presently a Phase-I development and is not claimed to be a clinically validated product. The recovery and recommendation components require validation using real clinical rehabilitation datasets, while the exercise-monitoring component requires further testing across different body types, camera positions, lighting conditions and exercise environments. The security and privacy mechanisms also need to be strengthened as the system progresses because CAREX handles health-related

information. These limitations do not prevent the proposed architecture from serving as a foundation for further development, but they indicate that additional implementation, testing and clinical evaluation are required before the system can be considered for real-world healthcare deployment.

Overall, the CAREX implementation establishes a unified framework for personalized rehabilitation by combining patient-specific data, exercise monitoring, computer vision, machine-learning-based recovery analysis, reminders, conversational assistance and healthcare-professional supervision. The system is designed to reduce dependence on periodic monitoring by providing continuous digital support while retaining professional oversight. Future implementation can extend the existing framework through wearable integration, advanced computer vision, fall detection, multilingual assistance, improved predictive analytics and larger clinical datasets, allowing CAREX to evolve into a more comprehensive rehabilitation platform.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

CAREX brings personalised exercise planning, continuous progress monitoring, recovery prediction, and clinician supervision together within one platform, aiming to reduce the reliance conventional rehabilitation places on in-person visits and self-reported progress while keeping a clinician in charge of every treatment decision. The system integrates patient-specific information, rehabilitation activities, AI-assisted analysis, progress tracking, medication and appointment reminders, and a conversational assistant to provide continuous support during home-based rehabilitation.

Reviewing the literature shows that telerehabilitation, AI-based exercise recommendation, and machine-learning outcome prediction have each been studied on their own, with encouraging but methodologically limited results. CAREX combines these components into a single adaptive environment rather than proposing a new algorithm for any one of them. The architecture provides a foundation for personalised and data-driven rehabilitation while maintaining healthcare-professional involvement in the decision-making process.

The current system is a starting point rather than a clinically validated product. Further development is required to validate the recommendation and prediction models using real clinical data, evaluate the platform with patients and healthcare professionals, and strengthen the security and privacy mechanisms required for handling health-related information.

6.2 FUTURE ENHANCEMENTS

The future development of CAREX can focus on extending its capabilities for more accurate, accessible, and effective rehabilitation support. The major areas of future enhancement include:

- **IoT-Based Wearable Integration:** Integrate an ESP32 and MPU6050-based wearable device to capture real-time movement and motion data during rehabilitation exercises.
- **Real-Time Exercise Monitoring:** Combine wearable sensor data with computer vision to improve exercise recognition, movement tracking, and posture assessment.
- **Fall and Abnormal-Movement Detection:** Detect sudden falls or abnormal movement patterns and trigger SOS alerts to caregivers or healthcare professionals.
- **Advanced Computer Vision:** Enhance pose estimation with joint-angle, range-of-motion, repetition, and exercise-correctness analysis for more detailed rehabilitation monitoring.
- **Doctor/Clinician Portal:** Enable healthcare professionals to remotely monitor patient progress, review AI-generated insights, and update rehabilitation plans.
- **Voice and Multilingual Assistant:** Provide hands-free exercise instructions, real-time feedback, and multilingual guidance to improve accessibility.
- **Predictive Recovery Analytics:** Improve machine-learning models using larger rehabilitation datasets to predict recovery progress, potential risks, and required changes in exercise intensity.
- **Gamification and Patient Engagement:** Introduce recovery milestones, badges, goals, and progress-based rewards to encourage consistent participation in rehabilitation.

These enhancements can help CAREX evolve into a more comprehensive rehabilitation platform while continuing to maintain healthcare-professional involvement in patient care.

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